

Answers

1.D	2.C	3.D	4.B	5.D	6.C	7.C	8.D
9.D	10.D	11.A	12.A	13.D	14.D	15.B	16.B
17.A	18.6	19.D	20.4	21.B	22.A	23.C	24.A
25.D	26.D	27.A	28.C	29.C	30.45	31.A	32.C
33.B	34.A	35.C	36.D	37.A	38.D	39.B	40.B
41.B	42.D	43.A	44.B	45.B	46.B	47.B	48.C
49.B	50.A	51.B	52.D	53.C	54.A	55.B	56.B
57.A	58.C	59.B	60.C	61.B	62.D	63.A	64.30
65.A	66.A	67.1	68.3	69.C	70.C	71.A	72.B
73.5	74.A	75.5	76.A	77.A	78.40	79.2	80.D
81.C	82.A	83.4	84.10	85.B	86.B	87.C	88.B
89.B	90.C	91.C	92.C	93.2	94.D	95.A	96.A
97.D	98.C	99.A	100.D	101.A	102.B	103.B	104.D
105.B	106.B	107.A	108.D	109.B	110.B	111.A	112.C
113.C	114.A	115.B	116.B	117.C	118.B	119.B	120.D
121.B	122.B	123.C	124.C	125.A	126.C	127.C	128.C
129.C	130.B	131.D	132.D	133.C	134.D	135.D	136.D
137.A	138.C	139.C	140.A	141.C	142.B	143.34	144.42
145.A	146.B	147.B	148.B	149.D	150.A	151.150	152.B
153.C	154.B	155.A	156.A	157.D	158.C	159.A	160.B
161.D	162.B	163.A					

Explanations

1. D

Year	Plywood in Rs/Tonne	Plywood in Rs/meter cube	Saw Timber in Rs/Tonne	Saw Timber in Rs/meter cube	Difference
1989	4	2.8	12	9.6	6.8
1990	5	3.5	10	8	4.5
1991	4	2.8	13	10.4	7.6
1992	6	4.2	15	12	7.8

From the above diagram, we can say that difference is highest in 1992.

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2. C

In this case the average realization will be-

$$\frac{(4 \times 5.26 \times 1.05) + (3 \times 14.28 \times 1.01) + (3 \times 20 \times 1.1)}{4 + 3 + 3} \approx 13.15$$

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3. D

In the diagram, we can see the sudden increase in MAY, and there was no such increase again during the given period.

Hence, May is the answer.

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4. B

There were 30 naya mixture in 97' which will also work in 98' ,thus we have 50 new mixtures in 98' . Now in 99' 20% of 30 i.e. 6 are thrown in junk . So in 99' we have $24 + 50 = 74$ naya mixtures of previous years. So naya mixture grinder purchased in 99' is $124 - 74 = 50$. Hence option b.

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5. D

Since X axis gives expenditure we have to see if average is as right as possible. Average expenditure of dubey family is rightmost. Hence, option D.

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6. C

Since the Y-axis gives the income, to find minimum average income see if any family has income in different months as below as possible in the graph.

From the graph we can see that coomar has lowest average income. Hence option C.

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7. C

Let the GDP of china and India be c and i respectively. According to given condition . We know that (Foreign inflows)china = $10 \times$ (Foreign inflows)India. And Foreign inflows = $FEI \times GDP$. So we get , $4.8 \times c = 10 \times 0.72 \times i$; which is $c = 1.5 \times i$. Hence , China's GDP in 1998 was 50% higher than that of India.

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8. D

The given values in the graph are ratio of FEI to GDP un % . So if entire value increase we can conclude that there can be increase in numerator and decrease in value of denominator and if entire value decrease we can conclude that there can be decrease in numerator and increase in value of denominator. hence only options 2 and 5 comply to these .

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9. D

Since we do not know the distribution of households in each region, we cannot determine the average income.

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10. D

In this question the total population does not change much in the whole span. So the per capita food grain production will be maximum when the total foodgrain production is maximum. It happens in 1995.

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11. A

The number of man-hours spent in coding = $425(\text{offshore}) + 100(\text{onsite}) = 525$

The number of man-hours spent in offshore design and offshore coding together = $100 + 425 = 525$

So, option a) is the correct answer

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12. A

The answer to this question is provided in the table.

The average income of middle income category is 30,000

The average income of upper-middle income category is 50,000 and

The average income of high income category is 75,000

Hence, the correct answer is 75,000

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13. D

Profitability in the year 1994-95 is $2.5/100 = 2.5\%$

Profitability in the year 1995-96 is $4.5/250 = 1.8\%$

Profitability in the year 1996-97 is $6/300 = 2\%$

Profitability in the year 1997-98 is $8.5/290 = 2.93\%$

Profitability in the year 1998-99 is $12/680 = 1.76\%$

Profitability increased/decreased in some years, hence the first three options are wrong.

So, the correct answer is option (d)

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14. D

Percentage growth in net profit in the year 1995-96 is $4.5/2.5 - 1 = 80\%$

Percentage growth in net profit in the year 1996-97 is $6/4.5 - 1 = 33.33\%$

Percentage growth in net profit in the year 1997-98 is $8.5/6 - 1 = 41.67\%$

Percentage growth in net profit in the year 1998-99 is $12/8.5 - 1 = 41.17\%$

Hence, the percentage growth in net profit is maximum in the year 1995-96

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15. B

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.

We know that the sales in April is 40.

Let the sales in May be $40+x$ and the sales in June be $40+2x$.

We know that the total sales in Q2 is 150.

$$\Rightarrow 40 + 40 + x + 40 + 2x = 150$$

$$3x = 30$$

$$x = 10$$

Therefore, sales in May 2016 = $40 + 10 = 50$

Sales in June 2016 = $40 + 20 = 60$

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic

progression.

Sales in October = 100

Sales in Q4 = 360

Let the sales in November be $100+y$ and the sales in December be $100+2y$.

$$100 + 100 + y + 100 + 2y = 360$$

$$300 + 3y = 360$$

$$\Rightarrow y = 20$$

Sales in November 2016 = 120 and Sales in December 2016 = 140

Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016

$$= 80 + 60 + 100$$

$$= 240$$

Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

$$= 75 + 120 + 55$$

$$= 250$$

Sales in Q1 of 2017 = $120 + 100 + 160 = 380$

Sales in Q2 of 2017 = $65 + 75 + 60 = 200$

We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be 'a'.

$$60 + 70 + a = 220$$

$$\Rightarrow a = 90$$

Sales in August 2017 = 90

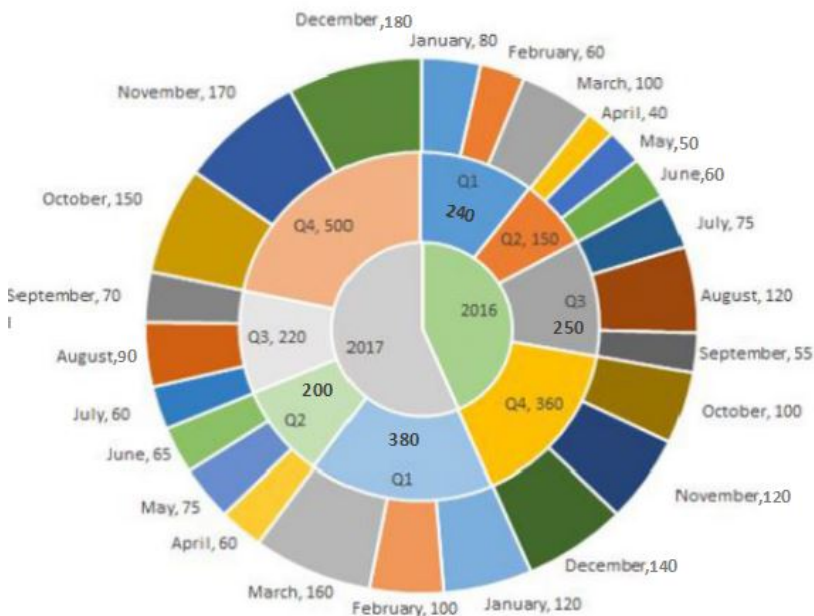
We know that sales in Q4 of 2017 = 500

Let the sales in December of 2017 be 'd'.

$$150 + 170 + d = 500$$

$$\Rightarrow d = 180$$

Sales in December 2017 = 180



March of 2017:

Sales in March of 2017 = 160

Sales in February of 2017 = 100

$$\% \text{ increase} = 60/100 = 60\%$$

October of 2017:

Sales in October of 2017 = 150
Sales in September of 2017 = 70
As we can see, the sales has increased by more than 100%.

March of 2016:
Sales in March of 2016 = 100
Sales in February of 2016 = 60
% increase in sales is less than 100%.

October of 2016:
Sales in October of 2016 = 100
Sales in September of 2016 = 55
% increase is less than 100%

As we can see, the percentage increase in sale as compared to the previous month was highest in October of 2017 among the given options. Therefore, option B is the right answer.

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16. **B**

As it is given that profit = sales - cost

So profit for september = $\frac{3}{36} \times 100 = 8\%$ (Nearly)

profit for march = $\frac{3}{30} \times 100 = 10\%$ (nearly)

profit for july = $\frac{2}{35} \times 100 < 10\%$

profit for may = $\frac{2}{36} \times 100 < 10\%$

So highest percentage profit is for march month.

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17. **A**

Simple calculations show difference in values of sales for karnataka over 2 successive pair of years - 1999-2000 and 2000-2001 - same.

The increase in 1999-2000 = 4839 - 4265 = 574

The increase in 2000-2001 = 5413 - 4839 = 574

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18. **6**

The frequency distribution is:

S: ~~3~~,3,3,4,4,4,5,5,6,7

F: ~~1~~,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

or

S: 3,3,3,4,4,4,5,5,6,7

F: ~~1~~,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

Zooma(Z) has a total score of 17 (comes under happy category), and other 2 countries, which are categorized as happy, have the highest score in exactly one parameter.

Suppose the other two countries are P and Q

Z have two possibilities for S, F, C : (6,7,4) & (6,5,6)

All the other cases are negated because "All the 3 countries, which are categorised as happy, have the highest score in exactly one parameter."

For Example : 7,7,3 is not possible because 7 being the highest score is there in two parameters.

So, it scored 6 in S in both the cases.

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19. D

The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

The highest cases are registered in West Bengal and Delhi.

The total number of IPC crimes = 63-64

The total number of SLL crimes = 520+35-36 = 555-556

Hence the ratio = (63-64)/(555-556) = 0.11 (Approximately) = 1:9

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20. 4

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$$I + II + III = 32 \dots (1)$$

$$I + 2II + 3III = 43 \dots (2)$$

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since $III=1$ we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is $7(\text{only two of them}) + 1(\text{USA}) = 8$, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries in ROW visited by both Nitesh and Samantha is $3+1=4$.

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21. B

Total tonnes of transportation by both rail and road is about $31 \times 12 / 100 = 3.72$ million tonnes and total cost incurred $18 \times 30 / 100 = 5.4$.

Hence required value is $5.4 / 3.72$ which is about 1.5.

Hence option B.

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22. A

Cheapest mode of transport will be the one which will have highest transport volume and comparatively lowest cost. We can figure out from the graph that Road have wide gap with very less costs.

For Road, Cost = $6/22$

For Rail, Cost = $12/9$

For Pipeline, = $65/49$

For Ship, Cost = $10/9$

Lowest cost is for road.

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23. C

Total increase in 4 years = $8640 - 5290 = 3350$

% increase = $\frac{3350}{8640} = 63.3\%$

Average per-year growth = $63.3/4 = 15.9\%$

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24. A

To have highest amount of savings, income should be high and expenditure should be low. For ahuja family the gap is highest. Hence they have highest amount of savings

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25. D

The profit per employee will be hishest when the ratio of profit to the number of employees is the highest. It happens in march and the value is $\frac{33-30}{12} = \frac{1}{4}$

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26. D

From the graph we can clearly make out that for dubey family in both the months income is nearly equal to expenditure, thus hey have lowest savings.

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27. A

We know that average of the particular area would increase by 1 each year if there is no change in any member of the group.

So, now in marketing, ob and om, we can see increase in average by 1 in 3 successive years.

But in Finance we see that average decreases 2 times, once when a new teacher is recruited and other time when the 60 year old retires.

Hence option A.

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28. C

Percentage of students more in marketing jobs than in finance jobs are 14%, 31%, 20%, 18% and 0% in 1992, 1993, 1994, 1995 and 1996 respectively.

=> Number of people = $14 \times 800 / 100 + 31 \times 650 / 100 + 20 \times 1100 / 100 + 18 \times 1200 / 100 = 112 + 202 + 220 + 216 = 750$

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29. C

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males in the age group of 30-40(in 1980) alive till 2010 = 90

The total number of females in the age group of 30-40(in 1980) alive till 2010 = 100

Thus, the total number of people in the age group of 30-40(in 1980) alive till 2010 = 90 + 100 = 190

Thus, the correct option is C.

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30.45

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to be rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

D was rewarded 45 stars in total.

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31. **A**

Percentage increase from 2006 to 2007 is $120/3.8 = 31.5\%$. approximately

Now if the same % increase is considered over period 2007-2008, we get . $31.5 * 500 = x-500$.

X is 657 approximately.

So ,the difference between the estimated subscription in Europe in 2008 and what it would have been if it were computed using the percentage growth rate of 2007 (over 2006), is closest to: $657-600 = 57$. Hence option A.

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32. **C**

Growth in 2007 $\Rightarrow \frac{500-380}{380} * 100 = 31.58$

Growth in 2005 $\Rightarrow \frac{280-190}{190} * 100 = 47.37\%$

Percentage change = $\frac{47.37-31.58}{47.37} * 100 = 35\%$ approximately.

$\Rightarrow 35\%$ is the answer

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33. **B**

Since in 94 total production was 50% more than 89 , so total production = 150. Also we know that total production in given 4 sectors in 94 = 76.8. Hence other than the 4 sectors total production is $150 - 76.8 = 73.2$. Hence the percentage increase in production between 1989 and 1994 in sectors other than the four listed above is: $(73.2-40) / 40$ is approx 87.5% . Hence option B.

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34. **A**

Let first approximately calculate production in-the four sectors, manufacturing, mining & quarrying, electricity and chemicals in 1994 we get , 25.4 , 18 , 14.55 , 18.85 respectively . Hence overall growth = 76.8. Now The percentage increase of production in-the four sectors, taken together, in 1994, relative to 1989, is approximately = $(76.8-60)/60$ which is approx. 25 % . Hence option A.

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35. **C**

From the graph it is clearly visible that Electricity had the highest growth during the period 1989 and 1998. Hence option C.

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36. **D**

We cant find absolute value of FEI as GDP value is not given . Hence option D.

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37. **A**

From the given data we can clearly notice that FEI for India decreased (changed) by greatest percentage.

For India the value = $(1.71 - 0.72) / 1.71 = 57.894\%$

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38. **D**

The overall revenue has increased by a ratio of $\frac{49639}{29870} = 1.67$.

So, the state whose revenue ratio increases by more than 1.67 is the answer.

Among the given options, only AP's sales tax revenue ratio has increased.

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39. **B**

We have to consider the ranking of that state that doesn't change throughout or changes only once.

The rankings of the states are as follows:

	1996 – 97	1997 – 98	1998 – 99	1999 – 00	2000 – 01
West Bengal	7	7	7	7	7
Uttar Pradesh	6	5	5	5	4
Tamil Nadu	2	2	2	2	2
Maharashtra	1	1	1	1	1
Karnataka	5	6	6	6	6
Gujarat	3	3	4	4	5
Andhra Pradesh	4	4	3	3	3

So, the states West Bengal, Tamil Nau, Maharashtra, Karnataka and Andhra Pradesh do not change their rankings more than once. There are 5 states in total under the criteria.

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40. **B**

Turkey has a value of 16% and a quantity of 15%.

The average price in Euro per kilogram for Turkey is $(16 \times 5.760) / (15 \times 1.055)$ which is nearly equal to 5.6.

Hence option B.

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41. **B**

The country which has the highest average price would have low quality and high value when compared to other countries..

Switzerland fits the above description.

Hence, option B is the answer.

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42. **D**

Since 10 purana were disposed off in 1997, it means 50 new have been bought in 1995, which implies that 70 were already in use. The breakup of 70 purana cannot be determined. So the number of purana disposed in 2000 cannot be calculated.

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43. **A**

Now from B through F total work done is 81.7%. Dividing it in 5 we get 16.34. So difference in operation E is $28.6 - 16.34 = 12.3\%$. Hence there is a reduction of 12.3 .

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44. **B**

There are 3 steel and 2 cement companies with a turnover more than 1000 and profit exceeding 10% of turnover. hence there are 5 choices for the investor.

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45. **B**

There are in total 7 companies - 3 steels , 2 cement , 2 textile for which the profit is more than 10% of the turnover.

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Data Interpretation for CAT Questions (download pdf)

46. **B**

Seeta's growth percentage = $(57-50)/50 * 100\% = 14\%$

Geeta's growth percentage = $(61-49)/49 * 100\% = 24.5\%$

Ram's growth percentage = $(61-52)/52 * 100\% = 17.30\%$

Shyam's growth percentage = $(61 - 53.5)/53.5 * 100\% = 14.01\%$

So, Seeta's growth percentage is the least.

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47. **B**

The rate of growth is the slope of the line. In the second and third months, there is an increase in the slope of the line. After the third month, there is a fall in the slope of the line i.e., the rate of growth is declining. Therefore, option b) is the correct answer.

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48. **C**

Max growth rate can be calculated by finding percentage growth over previous year.

We can see a significant increase in the year 1999-2000.

On calculating the ratio, we get $\frac{10284}{8067} = 1.275$, which is more than any other year.

Hence, 1999-2000 is the answer.

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49. **B**

The raw materials used in 1991 was 55,000

The raw materials used in 1992 was 48,000 (decrease by 7,000)

The raw materials used in 1993 was 60,000 (increase by 12,000)

The raw materials used in 1994 was 65,000 (increase by 5,000)

The raw materials used in 1995 was 75,000 (increase by 10,000)

Hence, the increase in use of raw materials was maximum in 1993

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50. **A**

The population is increasing steadily, but the milk consumed decreased significantly in 1996.

Hence, the only two years we need to check for the least per capita milk consumption are the first year (1990) and 1996.

Milk consumed in 1990 is 5 million gallons.

Total population in 1990 is $34+35 = 69$ million.

Hence, per capita consumption of milk in 1990 is 0.072 gallon per person.

Milk consumed in 1996 is 6 million gallons.

Total population in 1996 is $40+42 = 82$ million.

Hence, per capita consumption of milk in 1996 is 0.073 gallon per person.

Hence, the year with the least per capita consumption of milk is 1990.

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Logical Reasoning for CAT Questions (download pdf)

51. **B**

Volatility for arhar is $(2300-1500)/1900 = 42\%$

Volatility for pepper is $(19500-17500)/18500 = 10.8\%$

Volatility for sugar is $(1500-1410)/1450 = 6.2\%$

Volatility for gold is $(4300-3800)/4150 = 12\%$

Hence, the highest volatility is around 40%

 [VIEW SOLUTION](#)

52. **D**

The profit percentage equals $25\% \times (2105/1700 - 1) + 25\% \times (19250/18500 - 1) + 25\% \times (1430/1440 - 1) + 25\% \times (3850/4250 - 1)$

This equals $25\% \times (23.82\% + 4.05\% - 0.69\% - 9.41\%) = 25\% \times 17.77 = 4.4\%$

The closest option to this is option (d)

 [VIEW SOLUTION](#)

53. **C**

Volatility for arhar is $(2300 - 1500)/1900 = 42\%$

Volatility for pepper is $(19500 - 17500)/18500 = 10.8\%$

Volatility for sugar is $(1500 - 1410)/1450 = 6.2\%$

Volatility for gold is $(4300 - 3800)/4050 = 12\%$

Hence, the volatility is least for sugar.

 [VIEW SOLUTION](#)

54. **A**

Price change of arhar = $(2105 - 1700)/1700 = 23.82\%$

Price change of pepper = $(19250 - 18500)/18500 = 4.05\%$

Price change of sugar = $(1430 - 1440)/1430 = -0.69\%$

Price change of gold = $(3850 - 4250)/3850 = -9.41\%$

Hence, the price change is maximum for arhar.

 [VIEW SOLUTION](#)

55. **B**

In both imports and exports, India did the least amount of trade in 1997-98 with K (both equal to 1%).

Hence, India had the lowest total trade with K in the year 1997-98.

The amount of Indian exports equals $1\% \times 33979 = 340$ millions

 [VIDEO SOLUTION](#)

Quantitative Aptitude for CAT Questions (download pdf)

56. **B**

Profitability in the year 1994-95 is $2.5/100 = 2.5\%$

Profitability in the year 1995-96 is $4.5/250 = 1.8\%$

Profitability in the year 1996-97 is $6/300 = 2\%$

Profitability in the year 1997-98 is $8.5/290 = 2.93\%$

Profitability in the year 1998-99 is $12/680 = 1.76\%$

Hence, the profitability is maximum in the year 1997-98

 [VIEW SOLUTION](#)

57. **A**

Percentage growth in sales in the year 1995-96 is $250/100 - 1 = 150\%$

Percentage growth in sales in the year 1996-97 is $300/250 - 1 = 20\%$

Percentage growth in sales in the year 1997-98 is $290/300 - 1 = -3.33\%$

Percentage growth in sales in the year 1998-99 is $680/290 - 1 = 134\%$

Hence, the percentage growth in sales is maximum in the year 1995-96

 [VIEW SOLUTION](#)

58. C

After an increase by 33%, the difference in the prices of one tonne of saw timber and one cubic metre of logs is given below according to the respective years.

The difference in 1987 is $15 - 10 \times 133.33\% = 1.67$

The difference in 1988 is $16 - 10 \times 133.33\% = 2.67$

The difference in 1989 is $17.5 - 12 \times 133.33\% = 1.5$

The difference in 1990 is $15 - 10 \times 133.33\% = 1.67$

The difference in 1991 is $17.5 - 12.5 \times 133.33\% = 0.833$

The difference in 1992 is $18 - 15 \times 133.33\% = -2$

The difference in 1993 is $20 - 20 \times 133.33\% = -6.67$

Hence, the difference is least in the year 1991.

 [VIEW SOLUTION](#)

59. B

The percentage increase in price of logs over the period of time equals $20/15 - 1 = 33.33\%$

The percentage increase in price of plywood over the period of time equals $7/3 - 1 = 133.33\%$

The percentage increase in price of sawtimber over the period of time equals $200/100 - 1 = 100\%$

Hence, the maximum percentage increase in price over the period of time happened with plywood.

 [VIEW SOLUTION](#)

60. C

The maximum percentage increase in price of logs happened in the year 1991 (compared to 1990) and equaled $18/15 - 1 = 20\%$

The maximum percentage increase in price of plywood happened in the year 1992 (compared to 1991) and equaled $6/4 - 1 = 50\%$

The maximum percentage increase in price of sawtimber happened in the year 1993 (compared to 1992) and equaled $20/15 - 1 = 33.33\%$

Hence, the correct answer is 50%

 [VIEW SOLUTION](#)

Know the CAT Percentile Required for IIM Calls

61. B

The answer is provided by the bar chart.

Look at the graph named North and the bar colored Orange.

This shows the percentage increase in the upper middle class households from the year 1987-88 to 1994-95 and it equals 200%

 [VIEW SOLUTION](#)

62. D

We don't know the number of households in the upper-middle income class for Northern region and hence can't calculate the total household income for the same.

 [VIEW SOLUTION](#)

63. A

For having the greatest percentage change, the slope of the line should be the highest. From the graph, we can see that slope is highest for A between Feb and March. So A is the correct answer.

 [VIEW SOLUTION](#)

64. 30

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

We are given that there are 250 females from age 20-30 in 1980 and in 2000 these females age are from 40-50 but only 175 are alive in 2000.

In 2000 there were 175 females from age 40-50. If we assume that out of these, 30 females were of age 48 years in 2000 and they died in 2005, then there are 30 females who died at the age of 53.

If we assume that out of the 175 females, 30 females were of age 42 years in 2000, and they died in 2005, then 30 females died at the age of 47. Now, if we assume that there are 15 females of age 42 and 15 females of age 48 in the year 2000, and they all died in 2005, then we have 15 females who died at the age of 47 and 15 females who died at the age of 53.

So we can see that there are many cases possible. We are given that there were 250 females aged 20-30 in 1980, and in 2010, these females ages are from 50-60, but only 145 are alive in 2010.

In 2010 there were 145 females from age 50-60. If we assume that out of these, 40 females were of age 58 years in 2010 and they died in 2015, then there are 40 females who died at the age of 63.

If we assume that out of the 145 females, 40 females are of age 52 years age in 2010, and they died in 2015, then 40 females died at the age of 57. Now, if we assume that there are 15 females of age 52 and 25 females of age 58 in the year 2010, and they all died in 2015, then we have 15 females who died at the age of 57 and 25 females who died at the age of 63.

So we can see that again, there are many cases possible. In the first case, the range of values possible is from 0 to 30. In the second case, the range of values possible is from 0 to 40. So in total, we get a range of possible values from 0 to 70.

Thus, only one possible value of this question is not possible.

 [VIDEO SOLUTION](#)

65. A

In order to find the trading partner with the highest trade deficit, we have to look for countries from whom India imported a lot but exported little.

The two countries which are important here are G and H. However, the exports to both are the same. But the imports from H are more than the imports from G.

So, the trading partner with whom India has the highest trade deficit is H.

The amount of trade deficit equals $23\% \times 40779 - 10\% \times 33979 = 5981$ mio ~ 6 bn

 VIDEO SOLUTION

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66. A

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.

We know that the sales in April is 40.

Let the sales in May be $40+x$ and the sales in June be $40+2x$.

We know that the total sales in Q2 is 150.

$$\Rightarrow 40 + 40 + x + 40 + 2x = 150$$

$$3x = 30$$

$$x = 10$$

Therefore, sales in May 2016 = $40 + 10 = 50$

Sales in June 2016 = $40 + 20 = 60$

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic progression.

Sales in October = 100

Sales in Q4 = 360

Let the sales in November be $100+y$ and the sales in December be $100+2y$.

$$100 + 100 + y + 100 + 2y = 360$$

$$300 + 3y = 360$$

$$\Rightarrow y = 20$$

Sales in November 2016 = 120 and Sales in December 2016 = 140

Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016

$$= 80 + 60 + 100$$

$$= 240$$

Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

$$= 75 + 120 + 55$$

$$= 250$$

Sales in Q1 of 2017 = $120 + 100 + 160 = 380$

Sales in Q2 of 2017 = $65 + 75 + 60 = 200$

We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be 'a'.

$$60 + 70 + a = 220$$

$$\Rightarrow a = 90$$

Sales in August 2017 = 90

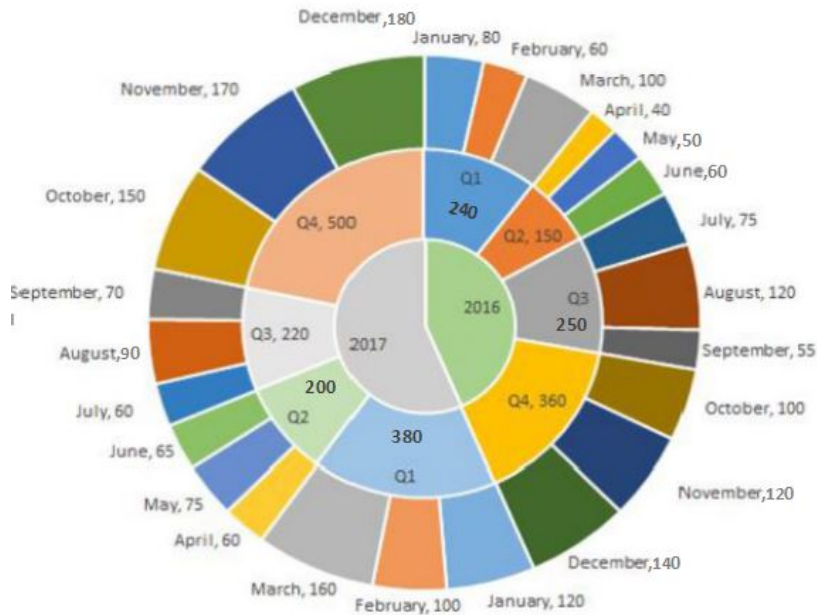
We know that sales in Q4 of 2017 = 500

Let the sales in December of 2017 be 'd'.

$$150 + 170 + d = 500$$

$$\Rightarrow d = 180$$

Sales in December 2017 = 180



Q2 of 2017:

Sales in Q2 of 2017 = 200

Sales in Q1 of 2017 = 380

$$\% \text{ decrease} = 180/380$$

Q4 of 2017:

We can eliminate this option since the sales has increased in Q4 of 2017 as compared to the previous quarter.

Q2 of 2016:

Sales in Q2 of 2016 = 150

Sales in Q1 of 2016 = 240

$$\% \text{ decrease} = 90/240$$

Q1 of 2017:

Sales in Q1 of 2017 has increased as compared to sales in the previous quarter. We can eliminate this option as well.

180/380 is very close to 50%. 90/240 is closer to 33.33%. Therefore, option A is the right answer.

[VIDEO SOLUTION](#)

67.1

The frequency distribution is:

S: 3,3,3,4,4,4,5,5,6,7

F: 1,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

or

S: 3,3,3,4,4,4,5,5,6,7

F: 1,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

Given Amda and Cadella score is 7 each with identical in all parameters. So it can score either 3,1,3 in S,F,C respectively or 4,1,2 in S,F,C respectively. In both the cases, its score in F is 1.

VIDEO SOLUTION

68.3

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$$I + II + III = 32 \dots (1)$$

$$I + 2II + 3III = 43 \dots (2)$$

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since $III=1$ we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is $7(\text{only two of them}) + 1(\text{USA}) = 8$, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

How many countries in Asia were visited by at least one of Dheeraj, Samantha and Nitesh is 3.

[VIDEO SOLUTION](#)

69. C

As the average age for OM area decreases in 2001, we can infer that the new faculty member joined OM on 1st April 2001. As his age was 25 on that date, his age on 1st April 2003 would be $25 + 2 = 27$ years. Thus, the new faculty member would be 27 years old. Hence option C.

[VIEW SOLUTION](#)

70. C

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

As fixed cost went up by 40 units, the total fixed cost (when number of units is less than 34) is $50 + 40 = 90$

Let the minimum number of units that are needed to be produced to ensure that there is no loss be X.

So, $20X \leq 14X + 90$

Or $X \leq 15$

So, the least possible number of units to be produced to ensure no loss is 15.

[VIDEO SOLUTION](#)

71. A

The rate of growth is the slope of the line. During the third month, Geeta has a negative slope whereas all the others have a positive slope. So, the rate of growth of Geeta is the least. Option a) is the correct answer.

[VIEW SOLUTION](#)

72. B

The set's starting point is the conclusion that the temperature rises only when the AC is turned off. When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celsius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

From this alone, we can answer the first question:

The AC was turned off for only two instances between 11:01 pm and 1:59 am:
Once at 0:00 and once at 1:00

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

73. 5

The data can be tabulated as follows (approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

Rank of Delhi in IPC crimes category = 1, The rank of Karnataka and Maharashtra is 3(from table), then the rank of Goa can only be 2.

The rank of Telangana is 6 which has less IPC crimes than Kerala, which means the rank of Kerala can be less than or equal to 5.

Now, there are two states with 3 ranks, so there will be no rank 4, there can only be rank 5 which is Kerala.

[VIDEO SOLUTION](#)

74. A

The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

From the table, the rank of Tamilnadu in other crimes is 2. The states which are not in the table will have crimes less than Telangana(i.e 24-25)

From the table the rank of Pudducherry in other crimes is 3.

[VIDEO SOLUTION](#)

75. 5

The data can be tabulated as follows(approximately):

The data can be tabulated as follows(approximately):

States	IPC Crimes	SLL Crimes	Other Crimes	Total Crimes
Telangana	3-4	14-15	6-7	24-25
Puducherry	1	0	30	31
Kerala	7-8	15-16	10-11	33-34
Haryana	3-4	25-26	9-10	37-38
Maharashtra	15-16	35-36	5-6	55-56
Tamilnadu	2-3	25-26	35-36	62-64
Goa	25-26	35-36	18-19	80
Karnataka	15-16	48-49	25-26	91
Delhi	63-64	35-36	42-43	142-143
West Bengal	0	520	0	520

	IPC crimes	SLL crimes	OtherCrimes
Delhi	*	*	*
Goa	*	4	*
Haryana	8	6	*
Karnataka	3	2	*
Kerala	*	9	*
Maharashtra	3	4	8
Puducherry	13	29	*
Tamil Nadu	11	7	*
Telangana	6	9	8
WestBengal	17	*	16

The rank of Delhi in IPC crimes should be 1 because the states which are not in table cannot crime more than that of Telangana which is 24-25.

Similarly Delhi Rank in Other crimes will be 1.

Now in SLL crimes clearly West Bengal has rank 1. It is given that Karnataka has rank 2. The rank 3 can go to either Goa, Delhi and Maharashtra but Goa and Maharashtra already have rank 4. So Delhi will have rank 3. Also no state outside of the table can be ranked 3 in SLL crimes as maximum number of crime should be less than that of Telangana(24-25). Here the number of SLL crimes is 35-36.

Hence the sum of the ranks = $1+3+1=5$

[VIDEO SOLUTION](#)

Cracku CAT Success Stories

76. **A**

After 1991 each year there is a positive growth rate change for manufacturing sector. Hence highest level of production would be in the year 1998.

[VIEW SOLUTION](#)

77. **A**

The rate of growth is the slope of the line.

From the graphs, we can see that the slope of the line is maximum for Geeta (red line) in the first two months. So, option a) is the correct answer.

[VIEW SOLUTION](#)

78. **40**

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males between 20 and 30 years of age in 1980 who died in 2000 = 205

The total number of males between 20 and 30 years of age in 1980 who died in 2010 = 165

Thus, the total number of males between 20 and 30 years of age in 1980 who died in the period 2000 to 2010 = 205 - 165 = 40

 VIDEO SOLUTION

79.2

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$$I + II + III = 32 \dots (1)$$

$$I + 2II + 3III = 43 \dots (2)$$

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since $III=1$ we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of them is $7(\text{only two of them}) + 1(\text{USA}) = 8$, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries visited by only Nitesh is 2.

 VIDEO SOLUTION

80.D

Since one was born in 1947 and the other one in 1950 and have birthday on 20th November, on 1st April 2000 they will have age 49 and 52 respectively.

Average age of all three on 1st April 2000 = 49.33 $\Rightarrow$ Total age = $49.33 \times 3 = 148$

Hence, age of the third member on 1st April 2000 = $148 - 49 - 52 = 47$

Age of the third member on 1st April 2005 = $47 + 5 = 52$.

Hence option D.

[VIEW SOLUTION](#)

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81. C

The difference is maximum when one quantity is maximum and other quantity is minimum. The percentage increase in production of food grains is maximum in 1995 and the percentage production increase in production of milk is negative for 1995.

[VIEW SOLUTION](#)

82. A

Purana mixer-grinders were purchased in 1999 = $236 - 222 + \text{Disposed in 1999}$
 $= 236 - 222 + 20\%(\text{Purana mixer-grinders were purchased in 1997})$

$= 236 - 222 + 20\%(182 - 162 + 10) = 236 - 222 + 6 = 20$.

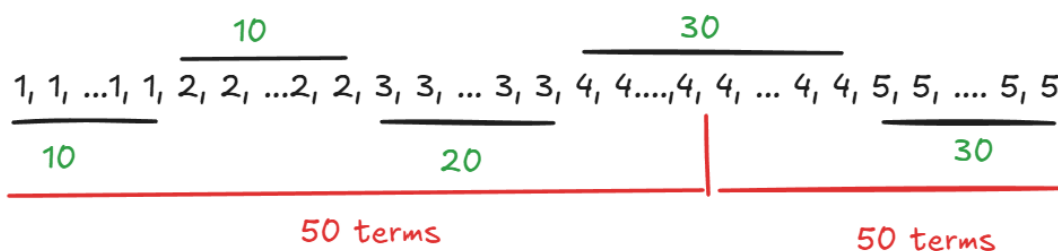
Hence option D.

[VIEW SOLUTION](#)

83. 4

As deduced in the previous question, there were 10 1 ratings, 10 2 ratings, 20 3 ratings, 30 4 ratings, and 30 5 ratings.

The median would be the 50th and 51st ratings' average when arranged in ascending/descending order. Both of which would be 4



Therefore, 4 is the correct answer

[VIDEO SOLUTION](#)

84. 10

The set's starting point is the conclusion that the temperature rises only when the AC is turned off.

When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celsius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree Celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

(Side note: We cannot consider the temperatures outside and inside at 23:00 and 2:00 since the time limit does not include those.)

The maximum difference between the elements of the outside and inside is maximum at 0:00 when the outside temperature is 36, and the inside temperature is 26.

Giving the maximum difference as 10

Therefore, 10 is the correct answer.

 VIDEO SOLUTION

85. B

Since the increase in imports and exports is at average, the exports and imports in the next 4 months will be half of their respective first eight months.

Exports in first 8 months of 1998-1999 is \$21436 so, in next 4 months it will be $21436/2 = 10718$

Total exports = $21346 + 10718 = 32154$.

Similarly, total imports = $28126 + 28126/2 = 42189$

Trade deficit = $42189 - 32064 = 10035$

Trade deficit in 1997-98 = $40779 - 33979 = 6800$.

Increase = $(10035 - 6800)/6800 = 47.5\%$

 VIDEO SOLUTION

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86. B

Till 34 units, the fixed cost is constant and the revenue increasing per every unit made.

So, to calculate the number of units that need to be produced to maximize the profit, we have to check only at two points: 34 units and 40 units.

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

If number of units produced is 34,

Fixed cost is 50

Variable cost is $14 \times 34 = 476$

Revenue is $20 \times 34 = 680$.

So, profit is $680 - 476 - 50 = 154$

If number of units produced is 40,

Fixed cost is 100

Variable cost is $14 \times 40 = 560$

Revenue is $20 \times 40 = 800$.

So, profit is $800 - 560 - 100 = 140$

So, the profit is maximized when the number of units produced is 34

 VIDEO SOLUTION

87. **C**

If the costs per tonne of transport by ship, air and road are represented by P, Q and R respectively.

$P = 10/9$

$Q = 7/11$

$R = 6/22$

We can see that $P > Q$ and $Q > R$ and $P > R$. Hence option C.

 VIEW SOLUTION

88. **B**

If suppose effort allocation is inter-changed between operations B and C, then C and D, and then D and E then operation E will have value of B. So arranging effort in E in ascending order we have company 4, company 5, company 3, ... Hence company 3 is ranked third. Hence option B.

 VIEW SOLUTION

89. **B**

Total number of hours needed for testing = $300 + 150 = 450$

New number of hours spend offshore for testing = $300/2 = 150$

So, required percentage = $150/450 \times 100\% = 33.33\%$

From the options, the closest answer is option b)

 VIEW SOLUTION

90. **C**

Total number of man-hours spent onsite = $75 + 100 + 150 = 325$

The sum of the estimated and actual effort for offshore design = 200 (approx)

The estimated man-hours of offshore coding = 425

The actual man-hours of offshore testing = 300 (approx)

Half of the man-hours of estimated offshore coding = 210 (approx)

So, option c) is the closest

91. C

Total number of man-hours needed for the entire work to be done = $100+75+425+100+300+150 = 1150$

Man-hours spent onsite = $75+100+150 = 325$

So, required percentage = $325/1150 * 100\% = 13/46 * 100\% = 28.26\% = 30\%$ approx

[VIEW SOLUTION](#)

92. C

There are 2 steel companies with turnover of more than 2000 and having profit less than 300.

[VIEW SOLUTION](#)

93. 2

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to be rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

M distributed among 3 bloggers, N among 2 bloggers, O among 1, P among 2, X among 1, Y among 3

Hence 2 surfers distribute their stars among 2 bloggers.

[VIDEO SOLUTION](#)

94. D

In 1993, price of logs = Rs. 20 per cubic metre.

volume sales of plywood, saw timber and logs were 40%, 30% and 30%

In 1993, price of plywood = Rs. 7 per tonne

In 1993, price of plywood = Rs. 20 per tonne

Weight of one cubic metre of saw dust and plywood both = 800 kg

1 tonne=1000kg

Price of 1000kg of plywood= Rs 7

Price of 800kg(1 m³)= 7 x 0.8 = Rs 5.6 per cubic metre.

Price of saw timber = 20 x 0.8 = Rs 16 per cubic metre.

We know that the sales volume of plywood, saw timber and logs are in the ratio 4 : 3 : 3.

Thus, the average realization must be the weighted avg of these three-

$$\frac{(4 \times 5.6) + (3 \times 16) + (3 \times 20)}{4 + 3 + 3}$$

$$= 13.1$$

= Rs. 13 (Approximately)

[VIEW SOLUTION](#)

95. A

To answer this question, let us calculate the highest percentage increase in market value of each share.

For share A, the highest percentage increase in market value is from Feb to March and equals $110/90 - 1 = 22.22\%$

For share B, the highest percentage increase in market value is from Jan to Feb and equals $75/70 - 1 = 7.14\%$

For share C, the highest percentage increase in market value is from Feb to March and equals $60/55 - 1 = 9.09\%$

For share D, the highest percentage increase in market value is from Jan to Feb and equals $50/40 - 1 = 25\%$

Hence, the required answer is February and the required share is share D

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96. A

Suppose there are 100 subscribers from Europe in the year 2003.

Out of which 60 are men and 40 are women.

According to given condition in the year 2010 no. of male subscribers would be $60 \times 1.05^7 = 84$ and no. of female subscribers would be $40 \times 1.1^7 = 78$.

So total equal to 162.

Hence % increase is $100 \times (162 - 100) / 100 = 62$.

Hence option A.

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97. D

Assume total base to be 100 (index = 100).

Contribution of manufacturing sector = $20\% = 20$ (in 1989).

In 1990, it has grown by 9%.

$\Rightarrow$ Contribution in 1990 = $1.09 \times 20 = 20 + 1.8 = 21.8$.

Hence, the growth of manufacturing sector in 1990 is 21.8. Similarly for other sectors the growth rates for mining 15.6, electricity 10.9 and chemicals is 16.2. So total overall growth is 64.5%.

Growth of manufacturing sector in 1991 is $21.8 - 0.01 \times 21.8 = 21.6$. Similarly for other sectors the growth rates for mining 15.7, electricity 11.88 and chemicals is 16.36. So total overall growth is 65.54%. Hence the growth rate change in 1991 is $(65.54 - 64.5) / 64.5 = 1.5\%$.

Hence option D.

[VIEW SOLUTION](#)

98. C

We know that the Excise duty is levied on the total value of liquor produced by the 5 distilleries. Since the excise duty remains constant, the excise duty will be in the same order as the order of the amount of the liquor produced by them.

Thus, the correct order is DCEBA.

[VIEW SOLUTION](#)

99. A

Loss from the transaction from Ram's standpoint is Sum of share prices of C and D - Share price of A.

So, loss from the transaction at the end of January is $(40+60) - 100 = 0$

So, loss from the transaction at the end of February is $(50+55) - 90 = 15$

So, loss from the transaction at the end of March is $(60+50) - 110 = 0$

So, loss from the transaction at the end of April is $(65+40) - 105 = 0$

So, loss from the transaction at the end of May is $(60+45) - 100 = 5$

So, loss from the transaction at the end of June is $(55+45) - 110 = 0$

So, the correct answer is 15

 [VIEW SOLUTION](#)

100. **D**

Loss from the transaction from Ram's standpoint is Sum of share prices of C and D - Share price of A.

So, loss from the transaction at the end of January is $(40+60) - 100 = 0$

So, loss from the transaction at the end of February is $(50+55) - 90 = 15$

So, loss from the transaction at the end of March is $(60+50) - 110 = 0$

So, loss from the transaction at the end of April is $(65+40) - 105 = 0$

So, loss from the transaction at the end of May is $(60+45) - 100 = 5$

So, loss from the transaction at the end of June is $(55+45) - 110 = 0$

So, the correct answer is February.

 [VIEW SOLUTION](#)

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101. **A**

To answer this question, let us calculate the highest increase in market value of each share.

For share A, the highest increase in market value is from Feb to March and equals $110 - 90 = 20$

For share B, the market value increases constantly over all the months and is equal to $75 - 70 = 5$

For share C, the highest increase in market value is from Feb to March and equals $60 - 55 = 5$

For share D, the highest increase in market value is from Jan to Feb and equals $50 - 40 = 10$

Hence, the correct answer is March and the required share is share A

 [VIEW SOLUTION](#)

102. **B**

S: 3,3,3,4,4,4,5,5,6,7

F: 1,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

Benga and Delma, two countries categorized as happy, are tied with the same total score.

The best numbers remaining are 7,5,6 which adds upto 18, If Benga scores 18, then Delma can't score 18.

Similarly both can't score 17 and 16. Both can score 15 and their distribution will be:

Benga: 7,5,3

Delma: 4,5,6 or 5,4,6

 [VIDEO SOLUTION](#)

103. **B**

S: 3,3,3,4,4,4,5,5,6,7

F: 1,1,2,3,3,4,5,5,5,7

C: 1,2,2,2,3,3,3,3,4,6

Given that Benga scores 16, and Delma scores 15.

The possibility is Benga: 5,5,6 and Delma: 7,5,3

If Benga's distribution is 7,3,6 then Delma can't score 15.

Strike off those numbers.

S: ~~3,3,3,4,4,4,5,5,6,7~~

F: ~~1,1,2,3,3,4,5,5,5,7~~

C: 1,2,2,2,3,3,3,3,4,6

We have to maximum number of countries with score 13. This score does not comes under the category of happy. So to score 13, the distribution can be 5,5,3. Hence, maximum 1 country is possible.

 VIDEO SOLUTION

104. D

The set's starting point is the conclusion that the temperature rises only when the AC is turned off.

When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celcius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

We can see that the AC was turned on at 0:30, 1:30 and 23:00.

We do not know about the status at 23:30

We are to count the times after 11:01 pm; hence, turning on the AC at 23:00 does not count.

Hence, the answer should be 2 or 3.

But in the final clue, we are given that POWER mode was used longer than REGULAR mode.

Since there are 4 periods of 30 minutes each, the combinations possible are POWER-3, REGULAR-1, and POWER-4, REGULAR-0

If the AC were on REGULAR mode at 23:00, it would have functioned for 1 hour, and the power mode would have also functioned for 1 hour (total), which would go against the final clue.

The following would be the cases.

	Case-1	Case - 2	Case - 3	Case - 4	Case - 5
23:00	Regular	Power	Power	Power	Power
23:30	Power	Regular	Power	Power	Power
0:30	Power	Power	Regular	Power	Power
1:30	Power	Power	Power	Regular	Power

But there were three instances of activation between 11:01 and 1:59

Hence, Option D would be the correct answer.

[▶ VIDEO SOLUTION](#)

105. **B**

%increase in profit in year 1995 = 16.66%

With this increment, profit in year 1996 = $70(1 + (16.66 \div 100)) = 82$ (nearly)

[▶ VIEW SOLUTION](#)



106. **B**

Number of Naya mixer-grinders were disposed off by the end of 1999 are 6 and at the end of 2000 are 10. So in total 16 naya mixer disposed off.

[▶ VIEW SOLUTION](#)

107. **A**

We are given that on day 3, a total of 100 ratings came in

The modes were 4 and 5, meaning that an equal number of 4 and 5 ratings came in; let it be $10b$ (since we are given that all ratings were non-zero multiples of 10)

Let's take the number of 1 and 2 ratings as $10a$ each, giving the number of 3 ratings as $20a$

Adding all of these up: $10a + 10a + 20a + 10b + 10b = 100$

$40a + 20b = 100$

$2a + b = 5$

The only integer combination for a and b , without them being zero, is a being 1 and b being 3 or $a=2$ and $b=1$

But taking b as 1 would make 3 as the mode rating, so we can not consider the latter case.

We are giving 10 - 1 ratings, 10 - 2 ratings, 20 - 3 ratings, 30 - 4 ratings, and 30 - 5 ratings.

The average would be $\frac{(10 \times 1) + (10 \times 2) + (20 \times 3) + (30 \times 4) + (30 \times 5)}{100} = \frac{360}{100} = 3.6$

Therefore, Option A is the correct answer.

[▶ VIDEO SOLUTION](#)

108. **D**

Every year, there was a comparative increase in both US and Europe subscriptions.

But in 2009, the increase in US subscriptions is very marginal and in Europe subscriptions is similar to that of any other year.

Hence, the percentage change in 2009 over 2008 would be more than any other year.

109. B

If she drives at 40 kmph, the fuel consumed = $200/40 \times 2.5 = 12.5$ liters

If she drives at 60 kmph, the fuel consumed = 13.33 liters

If she drives at 80 kmph, the fuel consumed = $200/80 \times 7.9 = 19.75$ liters

To minimize fuel consumption, Manasa has to decrease the speed.

110. B

Till 34 units, the fixed cost is constant and the revenue increasing per every unit made.

So, to calculate the number of units that need to be produced to maximize the profit, we have to check only at two points: 34 units and 45 units.

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

If number of units produced is 34,

Fixed cost is 50

Variable cost is $14 \times 34 = 476$

Revenue is $20 \times 34 = 680$.

So, profit is $680 - 476 - 50 = 154$

If number of units produced is 45,

Fixed cost is 100

Variable cost is $14 \times 45 = 630$

Revenue is $20 \times 45 = 900$.

So, profit is $900 - 630 - 100 = 170$

So, the profit per unit is maximized when the number of units produced is 34

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111. A

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males alive in 2000 = $180 + 205 + 160 + 100 = 645$

Thus, the number of dead males in 2000 = $1000 - 645 = 355$

Similarly, the total number of dead females in 2000 = $1000 - (210 + 175 + 150 + 120) = 1000 - 655 = 345$

Thus, the required ratio = $355 : 345 = 71 : 69$.

Thus, the correct option is A.

 VIDEO SOLUTION

112. C

The total number of male and female test cases in 1980 = 1000

Age group(In 1980)	Males alive in 1990	Males alive in 2000	Males alive in 2010	Males alive in 2020
10-20	230	180	150	140
20-30	235	205	165	125
30-40	210	160	90	50
40-50	190	100	25	5

Age group(In 1980)	Females alive in 1990	Females alive in 2000	Females alive in 2010	Females alive in 2020
10-20	240	210	160	100
20-30	225	175	145	105
30-40	190	150	100	60
40-50	220	120	30	18

The total number of males less than 30 years (in 1980) survived until 2020 = $140 + 125 = 265$

The total number of females less than 30 years (in 1980) survived until 2020 = $100 + 105 = 205$

Thus, The total number of people less than 30 years (in 1980) survived until 2020 = $205 + 265 = 470$

Thus, the correct option is C.

 VIDEO SOLUTION

113. C

The average age in the Finance area dropped twice - once due to the retirement of a member and once due to the joining of a new member. We see that the average age of the faculty is around 45-50. This number is closer to the retirement age of 60 than to the joining age of 25. Hence, the new joining will produce a greater drop in average age than the retirement. Hence, the new faculty member must have joined in 2002 and the old faculty member must have retired in 2001. Hence option C.

 VIEW SOLUTION

114. A

Average of numbers 100.5, 67, 141, 143.9 and 65 will be

$$100 + \frac{41+0.5+43.9-33-35}{5} = 103.48$$

 VIEW SOLUTION

115. **B**

Production rate in 1990 = 15.6 ; Production rate in 1991 = 15.7 ; Production rate in 1992 = 15.85 ; Production rate in 1993 = 15.46. So clearly we can see that production rate would be lowest in the year 1993. Hence option B.

[VIEW SOLUTION](#)



116. **B**

Ranking of UP changed 2 times which is highest among other states.

[VIEW SOLUTION](#)

117. **C**

The four major trading partners with India in 1997-98 are A, G, H and I.
From the numbers, we can ignore the others for this questions.

Total trade with A in the year 1997-98 is $9\% \times 40779 + 19\% \times 33979 = 10,126$
Total trade with G in the year 1997-98 is $19\% \times 40779 + 10\% \times 33979 = 11,146$
Total trade with H in the year 1997-98 is $23\% \times 40779 + 10\% \times 33979 = 12,777$
Total trade with I in the year 1997-98 is $14\% \times 40779 + 20\% \times 33979 = 12,505$
Hence, the correct option is block H which is OPEC

[VIDEO SOLUTION](#)

118. **B**

Decline occur in years 1985 and 1988 respectively.
In year 1985, decline was nearly 33%
Whether in year 1988 decline was nearly 54%
Hence answer will be B.

[VIEW SOLUTION](#)

119. **B**

Average simple annual growth of the distilleries are-

$$\begin{aligned} \text{A} &\rightarrow \frac{12.89 - 6.41}{2 \times 6.41} \times 100 = 50.54\% \\ \text{B} &\rightarrow \frac{12.07 - 3.15}{2 \times 3.15} \times 100 = 141.58\% \\ \text{C} &\rightarrow \frac{11.92 - 1.64}{2 \times 1.64} \times 100 = 313.41\% \\ \text{D} &\rightarrow \frac{5.79 - 1.05}{2 \times 1.05} \times 100 = 225.71\% \\ \text{E} &\rightarrow \frac{4.21 - 2.45}{2 \times 2.45} \times 100 = 35.91\% \end{aligned}$$

We can see that the highest growth rate was in C and lowest in E.
So, if E would have grown by 313.41% in 2 years, its supply would have been-

$$\frac{2.45 \times 313.41 \times 2}{100} = 15.11 \text{ lts}$$

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120. **D**

In the question we are only provided with the data of five states. We don't know the excise duty rates for other states. Thus, the answer cannot be determined.

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121. **B**

The difference in the excise duty as a percentage is least amongst AP and Maharashtra and it equals $(60-52)/52 * 100$

This equals $8/52 * 100 = 200/13 = 15.38\%$ which is approximately equal to 15%

[VIEW SOLUTION](#)

122. **B**

Ratio in 1991 = $\frac{10}{120} = 0.833$

Ratio in 1992 = $\frac{25}{125} = 0.2$

Ratio in 1993 = $\frac{20}{145} = 0.138$

Ratio in 1995 = $\frac{20}{160} = 0.125$

1992 has maximum profit per unit cost.

[VIEW SOLUTION](#)

123. **C**

Total Profit = $10 + 25 + 20 + 0 + 20 = 75$

Total Cost = $170 + 175 + 195 + 200 + 210 = 950$

% of Profit = $\frac{75}{950} = 8\%$

[VIEW SOLUTION](#)

124. **C**

Ratio in 1992 = $\frac{25}{50} = 0.5$

Ratio in 1993 = $\frac{25}{60} = 0.4167$

Ratio in 1994 = $\frac{30}{70} = 0.428$

Ratio in 1995 = $\frac{30}{80} = 0.375$

Hence, overheads, as a percentage of raw material, is maximum in 1992.

[VIEW SOLUTION](#)

125. **A**

Interest has remained constant over the given years attaining the value of 50-60 in all the year. Only in the year 1994, the value of interest is little less.

[VIEW SOLUTION](#)



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126. **C**

The profits of the company became zero only in one year (1994)

Hence, the change in profit is maximum in the period 1993-94 and equals 100%.

Note that there was a decrease in profits in other years also, but the maximum possible change is 100% and it was occurred in 1993-94

[VIEW SOLUTION](#)

127. **C**

Ratio of nutrient in milk(1 gallon) and foodgrains(1 tonne) is 3:2

There is a huge spike in the consumption of food grains in 1995.

Total nutrients consumed in 1995 = $30 \times 80 + 7 \times 120 = 3240$ million gms.

This is more than any other year.

=> 1995 is the answer

[VIEW SOLUTION](#)

128. **C**

The maximum availability of the nutrient will be either in 1994 or 1995. because the cumulative production of milk and food grains is maximum in these years.

Availability in 1994 = $(120 \times 8) + (25 \times 80) = 2960$

Availability in 1995 = $(6 \times 120) + (31 \times 80) = 3200$

Hence the availability is maximum in 1995.

[VIEW SOLUTION](#)

129. **C**

The consumption of milk is comparatively lesser than that of food grains and also the consumption of milk does not change by much during the given period.

Hence, the change in food grains is the key to estimate the calorie count.

We can see the spike in consumption of food grains in 1995. Hence, percapita consumption is highest in 1995.

[VIEW SOLUTION](#)

130. **B**

Number of students in 1994 passed out = 1100.

Number of students in 1994 from finance = 23% of 1100 = 23×11

Number of students in 1994 from software = 21% of 1100 = 21×11

Difference in salary = $12(23 \times 11 \times 7550 - 21 \times 11 \times 7050) = 3379200 = 33.8$ lakhs

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131. **D**

As we do not know the average salary for students who did not pursue Finance, Marketing or Software jobs, we cannot find the average salary of all the students in 1993.

Hence, a unique value cannot be determined.

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132. **D**

Finance salary increase = $9810 - 5450 = 4360$

$$\% \text{ increase} = \frac{4360}{5450} \times 100 = 80\%$$

[VIEW SOLUTION](#)

133. **C**

We can see that the end value is a little more than 15000 but not more than 16000.

So the increase in number of employees is between 5000 to 6000.

According to the options, it is 5800.

[VIEW SOLUTION](#)

134. **D**

This percentage is maximum in May which is equal to $\frac{36-32}{32} = 1/8$

[VIEW SOLUTION](#)

135. **D**

%increase in expenditure in year 1990 = above 7 but lower than 8

in year 1991 = near to 4

in year 1992 = above 8

in year 1993 = 8%

in year 1994 = lower than 4

in year 1995 = above 7 but lower than 8

Hence highest percentage increase is in year 1992

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136. **D**

percentage increase in profit in year 1990 = 25%

in 1991 = 20%

in 1992 = 33.33%

in 1993 = 25%

in 1994 = 20%

in 1995 = 16.66%

So highest %increase is in year 1992.

[VIEW SOLUTION](#)

137. **A**

Total expenditure = $102 + 110 + 115 + 125 + 135 + 140 + 150 = 877$

Total revenue = $120 + 130 + 145 + 165 + 185 + 200 + 220 = 1165$

$$\% \text{expenditure to revenue} = \frac{877}{1165} \times 100 = 75 \text{ (nearly)}$$

[VIEW SOLUTION](#)

138. C

The cumulative average rating at the end of day 3 would be $\frac{(3.1 \times 200) + (3.6 \times 100)}{200 + 100}$

[the cumulative average rating on day 2 = 3.1

Number of ratings received by day 2 = 200]

$$\frac{360 + 620}{300} = \frac{980}{300} = 3.266$$

The increase in the cumulative average from day 2 to day 3 can be calculated as $\frac{3.266 - 3.1}{3.1} \times 100 \approx 5.34$

Which aligns with the statement given in option C.

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

139. C

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2024 as 110x

This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

The first question asks about the number of others in 2024, which is 55x, or simply 55,000

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

140. A

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2024 as 110x

This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and $\frac{2}{3}$ of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

There are 15x, that is, 15,000 kids in 2023, of which we are given that 10,000 use one app.

So, 5,000 use multiple apps.

The percentage of kids using multiple apps in 2023 would hence be $\frac{5000}{15000} \times 100 = \frac{100}{3} = 33.33\%$

Therefore, Option A is the correct answer.

 VIDEO SOLUTION

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141.C

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2024 as 110x

This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and $\frac{2}{3}$ of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

In 2023, there were 20,000 elders; in 2024, there were 33,000 elders.

The increase in percentage would be $\frac{33000-20000}{20000} \times 100 = \frac{13}{20} \times 100 = 65\%$

Therefore, Option C is the correct answer.

[VIDEO SOLUTION](#)

142. B

Reading the bar graphs, we can find the distribution of the subscribers as:

	Kids	Elders	Others
2023	15%	20%	65%
2024	20%	30%	50%

Taking clue one into consideration, let's take the total number of subscribers in 2023 as 100x and in 2024 as 110x

This would give the split of subscribers as follows:

	Kids	Elders	Others	
2023	15x	20x	65x	100x
2024	22x	33x	55x	110x

Clue 2 says that half of 22x, which is 11x and 2/3 of 33x, which is 22x, used one app, given that in 2024, out of 55x, 33x used one app.

Clues 3 and 4 are to be used with each other.

Clue 4 says that out of 15x kids, 10,000 used one app, and out of 20x Elders, 15,000 used multiple apps.

Clue 3 says that the kids who used multiple apps (15x - 10000) were the same as Elders who used one app (20x - 15000)

Equating these two, we get:

$$15x - 10000 = 20x - 15000$$

$$5x = 5000$$

$$x = 1000$$

In clue 2, we were given that 33000 kids and elders out of 55000 kids and elders use one app, showing that 22000 use multiple apps. To minimize the total usage of multiple apps, we can assume that all 55000 other subscribers use one app.

Giving the percentage of multiple app users as: $\frac{22,000}{110,000} \times 100 = \frac{100}{5} = 20\%$

Therefore, Option B is the correct answer.

[VIDEO SOLUTION](#)

143. 34

The set's starting point is the conclusion that the temperature rises only when the AC is turned off.

When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celsius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degrees Celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

From the table, we can determine that the temperature outside at 1 am was 34 degrees Celsius. Therefore, 34 is the correct answer.

 VIDEO SOLUTION

144.42

The set's starting point is the conclusion that the temperature rises only when the AC is turned off. When the AC is turned off, the temperature rises linearly, reaching the temperature at the time it was turned off in one hour.

The temperature rises in two instances,

From 0:00 to 0:30, it rises from 26 to 31, so by 1:00, it would have reached 36 degrees, meaning that the temperature at 0:00 would have been 36 degrees Celsius.

From 1:00 to 1:30, it rises from 26 to 30, so by 2:00, it would have reached 34 degrees, meaning the temperature at 1:00 would have been 34 degrees Celsius.

Calculating either of these would have been enough as we have the starting temperature at 23:00 at 38 degree Celsius. Since the temperature decreases linearly, we can find the temperature at every instance using another data point.

Drawing out the temperature outside, we get:

Time	Temperature outside	Temperature inside	Mode of AC used
23:00	38	38	
23:30	37	32	
0:00	36	26	-
0:30	35	31	
1:00	34	26	-
1:30	33	30	
2:00	32	28	-

Extrapolating this linear chain of temperature drop (2 degrees drop every hour), we can see that the temperature at 22:00 would be 40 degrees Celsius, and the temperature at 21:00 or 9 pm would be 42 degrees Celsius.

Therefore, 42 is the correct answer.

 VIDEO SOLUTION

145.A

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to be rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

D received 5 stars from Y.

 VIDEO SOLUTION

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146. B

We can note down the data from the chart into the form of a table, that gives us,

Surfer/Blogger	A	B	C	D	Total
M	10	0			30
N	25	0			30
O	0	0			30
P	5	25			30
X	0	0			30
Y	5	20			30
Total	45	45	45	45	180

We are told that D receives more stars than C from Y. Considering Y has already given 25 stars, it will give 0 stars to C and 5 stars to D.

The only two surfers who have not given any stars to A or B is O and X, and these are the two surfers to give all of their stars to a single blogger.

We are also told that M gives different stars to the four bloggers,

Since he has already given 0 and 10, the remaining distinct stars should add up to 20. The only numbers that are remaining that add up to 20 are 5 and 15.

We know that X rewards one of C or D 30 stars and O rewards one of C or D 30 stars. Given that, M could not have rewarded D 15 stars, since Y rewarded D 5 stars, and D is also going to be rewarded 30 stars by O or X, and since the total is same for all, which is 45. This is not possible.

This means that, M rewarded C 15 stars and D 5 stars.

This gives us two cases,

Case-1

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	30	0	30
P	5	25	0	0	30
X	0	0	0	30	30
Y	5	20	0	5	30
Total	45	45	45	45	180

Case-2

Surfer/Blogger	A	B	C	D	Total
M	10	0	15	5	30
N	25	0	0	5	30
O	0	0	0	30	30
P	5	25	0	0	30
X	0	0	30	0	30
Y	5	20	0	5	30
Total	45	45	45	45	180

We can use these two cases to answer the questions,

There are two cases formed since we cannot identify for certain two whom did O reward the stars to, We can identify the number of stars rewarded by M to C.

Hence only statement 1 can be identified uniquely.

[VIDEO SOLUTION](#)

147. **B**

Since the increase in imports and exports is at average, the exports and imports in the next 4 months will be half of their respective first eight months.

Exports in the first 8 months of 1998-1999 is \$21436 so, in the next 4 months it will be $21436/2 = 10718$

Total exports = $21346 + 10718 = 32154$.

Although total imports have been increased the percentage share for all is decreased expect A and G. This means surely one of A and G has the highest increase.

Percentage growth in exports of country A is $(23\% \times 32154 - 19\% \times 33979) / 19\% \times 33979 = 14.5\%$

Percentage growth in exports of country G is $(12\% \times 32154 - 10\% \times 33979) / 10\% \times 33979 = 13.5\%$

Hence, the region to which Indian exports witnessed the highest percentage growth is region A i.e USA

[VIDEO SOLUTION](#)

148. **B**

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.

We know that the sales in April is 40.

Let the sales in May be $40+x$ and the sales in June be $40+2x$.

We know that the total sales in Q2 is 150.

$$\Rightarrow 40 + 40 + x + 40 + 2x = 150$$

$$3x = 30$$

$$x = 10$$

Therefore, sales in May 2016 = $40 + 10 = 50$

Sales in June 2016 = $40 + 20 = 60$

Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic progression.

Sales in October = 100

Sales in Q4 = 360

Let the sales in November be $100+y$ and the sales in December be $100+2y$.

$$100 + 100 + y + 100 + 2y = 360$$

$$300 + 3y = 360$$

$$\Rightarrow y = 20$$

Sales in November 2016 = 120 and Sales in December 2016 = 140

Sales in Q1 of 2016 = Sum of the sales in the months of January, February, and March 2016

$$= 80 + 60 + 100$$

$$= 240$$

Sales in Q3 of 2016 = Sum of the sales in the months of July, August, and September 2016

$$= 75 + 120 + 55$$

$$= 250$$

Sales in Q1 of 2017 = $120 + 100 + 160 = 380$

Sales in Q2 of 2017 = $65 + 75 + 60 = 200$

We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be 'a'.

$$60 + 70 + a = 220$$

$$\Rightarrow a = 90$$

Sales in August 2017 = 90

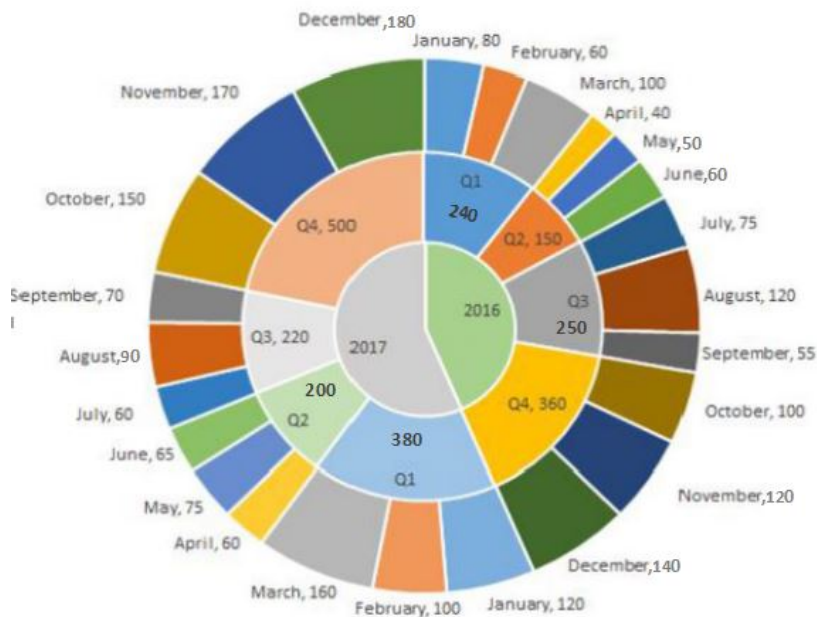
We know that sales in Q4 of 2017 = 500

Let the sales in December of 2017 be 'd'.

$$150 + 170 + d = 500$$

$$\Rightarrow d = 180$$

Sales in December 2017 = 180



Among the given 4 options, we have to find the quarter in which the increase in sale from the previous quarter was the highest.

Q2:

Sales in 2017 = 200

Sales in 2016 = 150

Q1:

Sales in 2017 = 380

Sales in 2016 = 240

Q3:

Sales in 2017 = 220

Sales in 2016 = 250

Q4:

Sales in 2017 = 500

Sales in 2016 = 360

We can eliminate Q3 since the sales has decreased.

Growth in Q2 sales = $50/150 = 1/3 = 33.33\%$

Growth in Q1 sales = $(380-240)/240 = 140/240 = 58.33\%$

Growth in Q4 sales = $(500-360)/360 = 140/360$

$140/240 > 140/360$

Therefore, Q1 has recorded the highest growth in sales and hence, option B is the right answer.

VIDEO SOLUTION

149. D

Writing down the data with us in a table form.

	Asia	Europe	Rest of World	Total
Dheeraj	3	7	1	11
Samantha	0	9	4	13
Nitesh	1	6	12	19
Total	4	22	17	43

But this includes countries which have been visited by one person, two people and three people. Essentially, there is an overlap of countries.

It is given to us that 32 distinct countries were visited among the three people,

Using that we can write down the equations,

$$I + II + III = 32 \dots (1)$$

$$I + 2II + 3III = 43 \dots (2)$$

Where I, II, and III denote the number of countries visited exactly by 1 of them, 2 of them and all of them.

We are told that USA(ROW) is the only country which was visited by all three people, that means $III = 1$

Subtracting Equation 1 from 2 we get, $II + 2III = 11$

And since $III=1$ we get $II = 9$

So, we are looking for 9 countries visited by exactly 2 people, and 22 countries visited by exactly 1 person.

Using the information from the problem set, we can note down the following,

Combination	Asia	EU	ROW	Total
Dheeraj				
Samantha				
Nitesh				
Dheeraj and Nitesh	1(China)			
Dheeraj and Samantha		1(France)		
Samantha and Nitesh				
All Three	0	0	1(USA)	
Total				32

We can see that Dheeraj visited only 1 country in ROW so every other value in that column has to be zero for wherever Dinesh is present. And Samantha has not visited any country in Asia so every value with Samantha present in Asia should be zero as well.

Combination	Asia	EU	ROW	Total
Dheeraj			0	
Samantha	0			
Nitesh				
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0			
All Three	0	0	1(USA)	
Total				32

We have the following distribution, we are looking for 9 countries visited by exactly 2 people. And we have identified 2 countries, that means, 7 were countries were visited by both Samantha and Nitesh only. And given that half the countries visited by both of them are in Europe, and total number of countries visited by both of

them is 7(only two of them)+1(USA)=8, they visited 4 countries in Europe together and 3 countries in ROW together. And using that information, we can fill in the rest of the table as well. Since we know the number of countries visited by each of them in a particular continent.

Combination	Asia	EU	ROW	Total
Dheeraj	2	6	0	8
Samantha	0	4	0	4
Nitesh	0	2	8	10
Dheeraj and Nitesh	1(China)	0	0	1
Dheeraj and Samantha	0	1(France)	0	1
Samantha and Nitesh	0	4	3	7
All Three	0	0	1(USA)	1
Total	3	17	12	32

Number of countries in EU visited by exactly 1 person is 12.

[VIDEO SOLUTION](#)

150. A

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

Let the minimum number of units that are needed to be produced to ensure that there is a profit of at least 50.

So, total revenue is $20X$ and total cost is $14X+50$.

So, Profit is $20X - (14X+50) = 6X-50 > 50$

Hence, $6X > 100$

Or $X > 16$

So, the least possible number of units to be produced is 17.

[VIDEO SOLUTION](#)

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151. 150

From the given chart, we can find the average rating on day 2:

$$\frac{(5 \times 1) + (10 \times 2) + (5 \times 3) + (20 \times 4) + (10 \times 5)}{5 + 10 + 5 + 20 + 10} = \frac{170}{50} = 3.4$$

We are given the cumulative average of day 1 and day 2 as 3.1, and the average at the end of day 1 is 3.

Let's take the number of ratings received on day 1 as x ; using this overall average and the average on day 2, we get the equation:

$$\frac{3x + (50 \times 3.4)}{x + 50} = 3.1$$

$$3x + 170 = 3.1x + 155$$

$$15 = 0.1x$$

$$x = 150$$

Therefore, the number of ratings received on day 1 is 150.

[VIDEO SOLUTION](#)

152. B

Average will be $= (120 + 130 + 145 + 165 + 185 + 200 + 220) \div 7 = 166$ (nearly)

153. C

We have been given details about the quarterly sales figures. Also, we have been given details about the sales figures every month. Some of the data are missing and some additional conditions have been given in the question. Let us try to complete the pie chart as much as possible with the data available to us.

It is known that the sales figures during the three months of the second quarter (April, May, June) of 2016 form an arithmetic progression.

We know that the sales in April is 40.

Let the sales in May be $40+x$ and the sales in June be $40+2x$.

We know that the total sales in Q2 is 150.

$$\Rightarrow 40 + 40 + x + 40 + 2x = 150$$

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$$x = 10$$

Therefore, sales in May 2016 = $40 + 10 = 50$

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Similarly, it has been given that the sales in October, November, and December 2016 form an arithmetic progression.

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$$100 + 100 + y + 100 + 2y = 360$$

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$$\Rightarrow y = 20$$

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 $= 80 + 60 + 100$
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We know that sales in Q3 of 2017 = 220

Let the sales in August of 2017 be 'a'.

$$60 + 70 + a = 220$$

$$\Rightarrow a = 90$$

Sales in August 2017 = 90

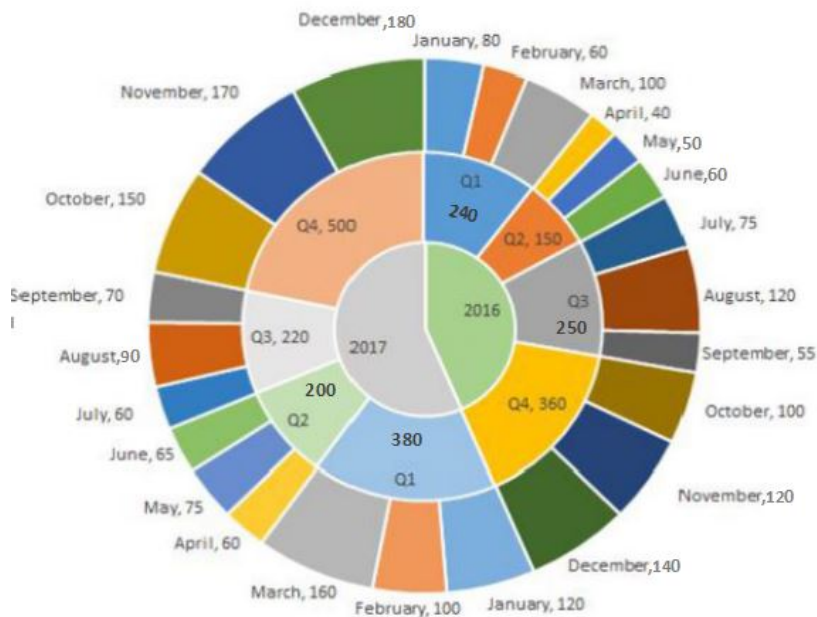
We know that sales in Q4 of 2017 = 500

Let the sales in December of 2017 be 'd'.

$$150 + 170 + d = 500$$

$$\Rightarrow d = 180$$

Sales in December 2017 = 180



Sales in December 2016 = 140

Sales in December 2017 = 180

Percentage change = $(180-140)/140 = 40/140 = 28.57\%$

Therefore, option C is the right answer.

[VIDEO SOLUTION](#)

154. **B**

At 60 kmph, time taken to reach the destination = $200/60 = 3.33$ hours.

Fuel consumed per hour at 60 kmph = 4 liters.

Therefore, total fuel consumed = $3.33 \times 4 = 13.33$ liters.

[VIEW SOLUTION](#)

155. **A**

Total working hours = 1150

So, we have to find the task where the number of working hours is approximately 575

Coding = $425 + 100 = 525$

Design = $100 + 75 = 175$

Offshore testing = 300

Offshore testing + design = $300 + 100 = 400$

So, option a) is the correct answer

[VIEW SOLUTION](#)

CAT Percentile Predictor

156. **A**

Total offshore work = $100+420+300 = 820$. If 50% of the offshore work were to be carried out onsite i.e 410 carried out onsite.

So new work carried out onsite is in design is $[80 + (100 \times 410)/(825)]$ which is around 130. Also new work carried out onsite is in coding is $[100 + (425 \times 410)/(825)]$ which is around 315.

And new work carried out onsite is in testing is $[150 + (300 \times 410)/(825)]$ which is around 300. Hence the amount of coding done is greater than that of testing.

 [VIEW SOLUTION](#)

157. **D**

Operations B, C and D have more effort weightage in companies 1,3,5,6.

Also we notice that among them operation E has highest already, and for E company 5 has already the highest percentage. So redistributing the share of E would increase further more. Hence option D.

 [VIEW SOLUTION](#)

158. **C**

Tamilnadu has maintained the ranking of second highest tax revenue throughout the five years.

Hence, option C is the answer.

 [VIEW SOLUTION](#)

159. **A**

Base value must be taken as 100 => 1985 is 100

$$1984 \Rightarrow \frac{100.5}{67} * 100 = 150$$

$$1986 \Rightarrow \frac{141}{67} * 100 = 211$$

Only option A has the same values => Option A is the answer

 [VIEW SOLUTION](#)

160. **B**

Percentage growth of all households South region showed the maximum increase in all households in the given period. $(100\% + 340\% + 425\% = 865\%)$

 [VIEW SOLUTION](#)

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161. **D**

Since we do not know the region-wise population of 1987-1988, we cannot find the increase in the number of households.

Hence, we cannot find the total income for the year 1995-96.

So the ratio cannot be determined.

 [VIEW SOLUTION](#)

162. **B**

The revenue of 50 units is 1000. So, the revenue per unit is 20.

The variable cost of 50 units is 700. So, the variable cost per unit is 14.

Let the minimum number of units that are needed to be produced to ensure that there is no loss be X.

$$\text{So, } 20X > 14X + 50$$

$$\text{Or } X > 8$$

So, the least possible number of units to be produced is 9.

 [VIDEO SOLUTION](#)

163. **A**

In order to find the country which has the least trade deficit with India, we have to look for countries whom India exported a lot and imported less.

The only two countries which fit the bill are countries A and I.

Trade deficit of A with India in 1997-98 is $9\% \times 40779 - 19\% \times 33979 = -2.78$ billion

Trade deficit of I with India in 1997-98 is $14\% \times 40779 - 20\% \times 33979 = -1.08$ billion

Hence, the country with the least trade deficit is country A or USA

 VIDEO SOLUTION

